

- 5 The variables x and y are such that $y = 0$ when $x = 0$ and

$$(x+1)y + (x+y+1)^3 = 1.$$

- (a) Show that $\frac{dy}{dx} = -\frac{3}{4}$ when $x = 0$. [3]

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- (b) Find the Maclaurin's series for y up to and including the term in x^2 . [7]

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